Nonce-Words Reveal Individual Differences in Korean RC Attachment Ambiguity

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Introduction. This study examines relative clause (RC) attachment patterns in Korean under conditions that minimize lexical-semantic information.

(1) ***The son of the doctor who was on the balcony*** won the lottery.

With sentences like (1), Korean users are generally known to favor high/non-local attachment (HA; Lim 2012, Lee 2025). Although cross-linguistic work has identified numerous factors (e.g. lexical biases, prosody) that influence attachment, it remains unclear to what extent observed language specific trends reflect syntactic parsing strategies. In this sense, recent work on English shows that when using plausible nonce words instead of fully interpretable nouns, the classic English LA profile disappears — allowing the disambiguation strategies of individual participants to surface more clearly (Lee & De Santo 2025). Here, we aim to explore this hypothesis in a language with significantly different structural properties and attachment profiles. Therefore, we conducted a similar experiment in Korean — a "HA" language with prenominal RCs – using nonce nouns that preserve morphosyntactic cues while eliminating real-world referential content. .

Materials. Following Lee & De Santo (2025), we conducted an offline forced-choice task. First, we build a set of phonotactically well-formed Korean nonce nouns by combining plausible stems with the morphemes *–kkwun* and *–cayngi*. Because these morphemes productively form nouns referring to human individuals characterized by particular activities or traits, their inclusion served to mark the nonce items as unambiguously human-referential, thereby enabling participants to interpret them as animate despite the items' lack of lexical familiarity. These nonce-nouns were then modified by pre-nominal RCs which could attach to either noun, using genitive morphology (*-uy*). Because Korean RCs precede the noun, the ambiguity appeared in the order RC NP2-*uy*(of) NP1, the structural counterpart of English NP1 of NP2 RC. We additionally manipulated RC length to explore whether prosodic grouping effects based on constituent length influence attachment (Hemforth et al., 2015). Short RCs consisted of two words, while long RCs were formed by adding two adjuncts (approximately 6–8 additional syllables). Twenty-four target sets and forty-eight fillers were distributed across two lists using a Latin-square design. Since Lee & De Santo (2025) report increased individual-level variability in attachment highlighted by the minimization of lexical-semantic influences, we are interested in exploring individual difference effects. In this sense, Swets et al. (2007) provide evidence for RC disambiguation strategies being influenced by participants' working memory capacity (WM). Therefore, we adapted their Spatial Span task, which participants completed before the forced-choice task.

Results. Pilot data (n=30) reveal a slight overall trend towards HA independently of Length (mean long: 0.59, sd: 0.49; mean short: 0.60, sd: 0.49). However, note how participants' preferences actually distribute bi-modally (Fig. 1 and Fig. 2) — unlike prior studies reporting strong HA biases in Korean with meaningful nouns. Regression analyses with Length and Z-transformed Span as fixed-effects and participant and sentence as random effects confirm no significant effect of Length ($p = 0.29$), nor any an effect of WM span ($p = 0.99$), nor any interaction ($p = 0.35$).

Conclusion. While preliminary, these results align with previous work showing that when using nonce-words in lieu of nouns with a full lexical semantics, individual trends in RC attachment prevail over language-level trends, in line with recent work casting doubts on a view of languages being simply "HA" or "LA". Such results then indicate that structural information alone does not sustain stable attachment outcomes, and suggest the use of nonce-words as an insightful paradigm to disentangle structural vs. lexical-semantic effects on parsing strategies during disambiguation. Although the individual variance found in our data could not be explained by the spatial span scores, further attention should be given to colleting diverse measures of cognitive resources both directly

related and independent from language processing.

Hemforth et al. 2015. Lingua. **Lee**. 2025. Language Research. **Lim**. 2012. Korean Journal of Linguistics. **Swets et al.** 2007. Journal of Experimental Psychology: General.

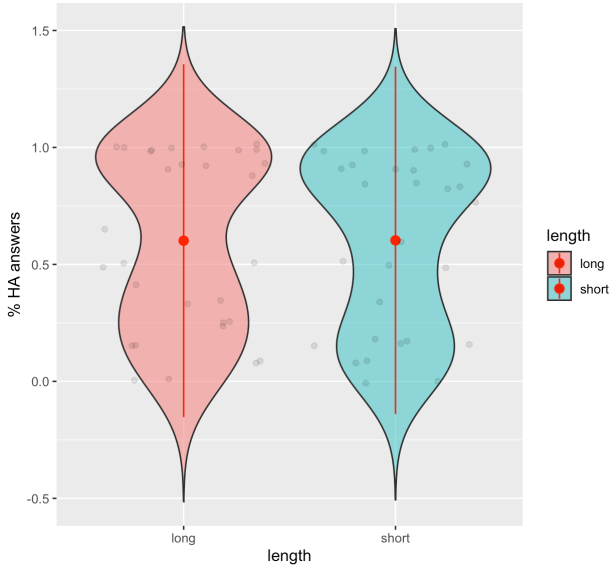

Figure 1: HA answer rate by Length. Red dot shows the Mean.

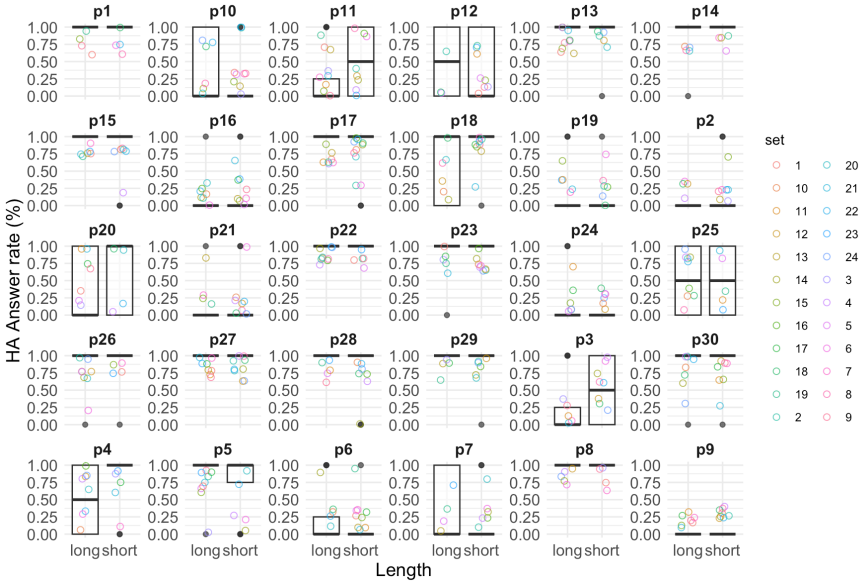

Figure 2: HA answer rate by Participant, Length, and Item Set.